2-1 $P_1$: $\vec{F}=500\cos 40^\circ\vec{i}+500\sin 40^\circ\vec{j}$

$P_2$: $\vec{F}=500\cos 10^\circ\vec{i}-500\sin 10^\circ\vec{j}$

$P_3$: $\vec{F}=250\sqrt{2}\vec{i}-250\sqrt{2}\vec{j}$

$\therefore \vec{F_R}=[500(\cos 40^\circ+\cos 10^\circ)+250\sqrt{2}]\vec{i}+[500(\sin 40^\circ-\sin 10^\circ)-250\sqrt{2}]\vec{j}$

$=1228.9\vec{i}-118.98\vec{j}$

$\therefore F_R=\sqrt{(1228.9)^2+(-118.98)^2}\approx 1234.7kN$

$\therefore \tan\theta=\left|\frac{F_{Ry}}{F_{Rx}}\right|=\frac{118.98}{1228.9}\approx 0.0968$

$\theta=\arctan(0.0968)\approx 5.54^\circ$

合力大小约1235N，水平向右，水平线向下偏约5.54°

3-1. 作用点在对角线为(-150, 200)

$\vec{F}$方向为(30, 40, 50)

$F_x=2000\times\frac{30}{\sqrt{50^2+40^2+50^2}}\approx 848.54N$

$F_y=2000\times\frac{40}{\sqrt{50^2+40^2+50^2}}\approx 1131.4N$

$F_z=2000\times\frac{50}{\sqrt{50^2+40^2+50^2}}\approx$

$M_z=xF_y-F_xy$ ∵单位为mm

$\therefore M=-150\times10^{-3}\times1131.4-848.54\times(200\times10^{-3})$

$=-339.42N\cdot m$

3-2. $M_z=F_z\cdot y-F_y\cdot z$ $M_y=F_x\cdot z-F_z\cdot x$ $M_z=xF_y-F_xy$

∵$F_y=0$ $z=0$

$M_x=F_z\cdot y$ $M_y=-F_z\cdot x$ $M_z=-F_xy$ $F_z=F\cdot\cos\theta$ $F_x=F\sin\theta$

作用点坐标：$(-b, a+l, 0)$

$\therefore M_x=F\cdot\cos\theta(a+l)$ $M_y=0$ $F\cdot\cos\theta\cdot(-b)$ $M_z=-F\sin\theta(a+l)$

$=-F\cos\theta(a+l)$ $M_y=-F\cos\theta b$

2-7. $F_1=F_2=400N$ $d_1=120\sqrt{3}+100$

$F_3=F_4=300N$ $d_2=0.48\sin 60^\circ+0.2\cos 60^\circ$

$d_3=100\sqrt{3}+200$

$d_4=0.48\sin 30^\circ-0.2\cos 30^\circ$

$\therefore M_{12}=d_{12}\times F_1=123.14N\cdot m$ 逆时针

$M_{34}=d_{34}\times F_3=123.96N\cdot m$ 逆时针

$M=M_{12}+M_{34}=247.1N\cdot m$

2-2. 建立坐标系



$\sum F_x=-P_1-P_2=-333kN$

$F_y=-F-Q=-8020kN$

$F_R=\sqrt{(8020)^2+(333)^2}=8027kN$

$\cos\alpha=\frac{F_{Rx}}{F_R}=-0.04148$

$\cos\beta=\frac{F_{Ry}}{F_R}=-0.99913$

$\Rightarrow\alpha=87.6^\circ$ $\beta=2.4^\circ$

$\sum M_O=\sum M_O(F_i)=M+F_x\cdot 2l+F_2\cdot 10.7$

$=10676kN\cdot m$ (逆时针)

作用线 $x=\frac{M_O}{F_R}=-1.331m$

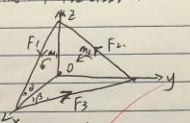
作用点在O点左侧1.331m

$M_A=M+M_O(\vec{P_1})+M_O(\vec{P_2})$

$=6103.5kN\cdot m$ 逆时针

$M_O\perp F_R$ $h=\frac{M_O}{F_R}=18.34m$

3. 四面体的三条棱OA、OB、OC相互垂直，且OA=OB=OC=a，在三角形ABC平面内的三条边上分别作用力$F_1$，$F_2$和$F_3$，且$F_1=F_2=\sqrt{2}F$，但$F_3=2\sqrt{2}F$，在OAC和OBC平面内分别作用力偶$M_1$和$M_2$，且$M_1=M_2=FR$。试将该力系向O点简化，并给出最终合成结果。



$F_x=F_1\cdot\cos\alpha-F_3\cdot\cos\beta$

$=\sqrt{2}F\cdot\frac{\sqrt{2}}{2}-2\sqrt{2}F\cdot\frac{\sqrt{2}}{2}=-F$

$F_{yz}=F_1\cdot\frac{\sqrt{2}}{2}+F_2\cdot\frac{\sqrt{2}}{2}=0$

$F_{yz}=F_3\cdot\frac{\sqrt{2}}{2}-F_2\cdot\frac{\sqrt{2}}{2}=F$

~~$\vec{M_3}=F_y\cdot a\vec{k}\times F_y\vec{j}=$~~ $\vec{F_R}=-F\vec{i}+F\vec{j}+0\vec{k}$

~~$M_{R3}=F_y\cdot a=Fa$~~ 在xoy平面上逆时针

合力对O点的矩：$\vec{M_O}(\vec{F})=\vec{r}\times\vec{F}$

$\vec{M_O}(\vec{F_1})=\vec{r_C}\times\vec{F_1}=\begin{vmatrix}\vec{i}&\vec{j}&\vec{k}\\0&0&a\\F&0&-F\end{vmatrix}=aF\vec{j}$

$\vec{M_O}(\vec{F_2})=\vec{r_C}\times\vec{F_2}=\begin{vmatrix}\vec{i}&\vec{j}&\vec{k}\\a&0&0\\0&-F&F\end{vmatrix}=aF\vec{k}$

$\vec{M_O}(\vec{F_3})=\vec{r_A}\times\vec{F_3}=\begin{vmatrix}\vec{i}&\vec{j}&\vec{k}\\a&0&0\\-F&F&0\end{vmatrix}=2aF\vec{k}$

$\vec{M_1}=Fa\vec{j}$ $\vec{M_2}=Fa\vec{i}$

$\therefore$ ~~$M_x=-2Fa$~~ $M_x=2Fa$ $M_y=2Fa$ $M_z=2Fa$

∵$F_R\neq 0$ $\vec{M_O}\neq 0$ 且$\vec{F_R}$与$\vec{M_O}$垂直 $\Rightarrow$

该合成结果为力螺旋。

